

How Volatility Affects Popular Strategies

The previous chapter addressed the calculation or interpretation of implied volatility, and how to relate it to historical volatility. Another, related topic that is important is how implied volatility affects a specific option strategy. Simplistically, one might think that the effect of a change in implied volatility on an option position would be a simple matter to discern; but in reality, most traders don't have a complete grasp of the ways that volatility affects option positions. In some cases, especially option spreads or more complex positions, one may not have an intuitive "picture" of how his position is going to be affected by a change in implied volatility. In this chapter, we'll attempt a relatively thorough review of how implied volatility changes affect most of the popular option strategies.

There are ways to use computer analysis to "draw" a picture of this volatility effect, of course, and that will be discussed momentarily. But an option strategist should have some idea of the general changes that a position will undergo if implied volatility changes. Before getting into the individual strategies, it is important that one understands some of the basics of the effect of volatility on an option's price.

VEGA

Technically speaking, the term that one uses to quantify the impact of volatility changes on the price of an option is called the *vega* of the option. In this chapter, the references will be to vega, but the emphasis here is on practicality, so the descriptions of how volatility affects option positions will be in plain English as well as in the more mathematical

realm of vega. Having said that, let's define vega so that it is understood for later use in the chapter.

Simply stated, vega is the amount by which an option's price changes when volatility changes by one percentage point.

Example: XYZ is selling at 50, and the July 50 call is trading at 7.25. Assume that there is no dividend, that short-term interest rates are 5%, and that July expiration is exactly three months away. With this information, one can determine that the implied volatility of the July 50 call is 70%. That's a fairly high number, so one can surmise that XYZ is a volatile stock. What would the option price be if implied volatility were to rise to 71%? Using a model, one can determine that the July 50 call would theoretically be worth 7.35 if that happened. Hence, the vega of this option is 0.10 (to two decimal places). That is, the option price increased by 10 cents, from 7.25 to 7.35, when volatility rose by one percentage point. (Note that "percentage point" here means a full point increase in volatility, from 70% to 71%.)

What if implied volatility had *decreased* instead? Once again, one can use the model to determine the change in the option price. In this case, using an implied volatility of 69% and keeping everything else the same, the option would then theoretically be worth 7.15—again, a 0.10 change in price (this time, a decrease in price).

This example points out an interesting and important aspect of how volatility affects a call option: *If implied volatility increases, the price of the option will increase, and if implied volatility decreases, the price of the option will decrease.* Thus, there is a *direct* relationship between an option's price and its implied volatility.

Mathematically speaking, vega is the partial derivative of the Black–Scholes model (or whatever model you're using to price options) with respect to volatility. In the above example, the vega of the July 50 call, with XYZ at 50, can be computed to be 0.098—very near the value of 0.10 that one arrived at by inspection.

Vega also has a direct relationship with the price of a put. That is, as implied volatility rises, the price of a put will rise as well.

Example: Using the same criteria as in the last example, suppose that XYZ is trading at 50, that July is three months away, that short-term interest rates are 5%, and that there is no dividend. In that case, the following theoretical put and call prices would apply at the stated implied volatilities:

Stock Price	July 50 call	July 50 put	Implied Volatility	Put's Vega
50	7.15	6.54	69%	0.10
	7.25	6.64	70%	0.10
	7.35	6.74	71%	0.10

Thus, the put's vega is 0.10, too—the same as the call's vega was.

In fact, it can be stated that a call and a put with the same terms have the same vega. To prove this, one need only refer to the arbitrage equation for a conversion. If the call increases in price and everything else remains equal—interest rates, stock price, and striking price—then the put price must increase by the same amount. A change in implied volatility will cause such a change in the call price, and a similar change in the put price. Hence, the vega of the put and the call must be the same.

It is also important to know how the vega changes as other factors change, particularly as the stock price changes, or as time changes. The following examples contain several tables that illustrate the behavior of vega in a typically fluctuating environment.

Example: In this case, let the stock price fluctuate while holding interest rate (5%), implied volatility (70%), time (3 months), dividends (0), and the strike price (50) constant. See Table 37-1.

In these cases, vega drops when the stock price does, too, but it remains fairly constant if the stock rises. It is interesting to note, though, that in the real world, when the underlying drops in price—especially if it does so quickly, in a panic mode—implied volatility can increase dramatically. Such an increase may be of great benefit to a call holder, serving to mitigate his losses, perhaps. This concept will be discussed further later in this chapter.

The above example assumed that the stock was making instantaneous changes in price. In reality, of course, time would be passing as well, and that affects the vega, too. Table 37-2 shows how the vega changes when time changes, all other factors being equal.

Example: In this example, the following items are held fixed: stock price (50), strike price (50), implied volatility (70%), risk-free interest rate (5%), and dividend (0). But now, *we let time fluctuate*.

Table 37-2 clearly shows that the passage of time results not only in a decreasing call

TABLE 37-1.

Stock Price	Implied Volatility July 50 Call Price	Theoretical Call Price	Vega
30	70%	0.47	0.028
40		2.62	0.073
50		7.25	0.098
60		14.07	0.092
70		22.35	0.091

TABLE 37-2.

Stock Price	Implied Volatility	Time Remaining	Theoretical Call Price	Vega
50	70%	One year	14.60	0.182
		Six months	10.32	0.135
		Three months	7.25	0.098
		Two months	5.87	0.080
		One month	4.16	0.058
		Two weeks	2.87	0.039
		One week	1.96	0.028
		One day	0.73	0.010

price, but in a decreasing vega as well. This makes sense, of course, since one cannot expect an increase in implied volatility to have much of an effect on a very short-term option—certainly not to the extent that it would affect a LEAPS option.

Some readers might be wondering how changes in implied volatility itself would affect the vega. This might be called the “vega of the vega,” although I’ve never actually heard it referred to in that manner. The next table explores that concept.

Example: Again, some factors will be kept constant—the stock price (50), the time to July expiration (3 months), the risk-free interest rate (5%), and the dividend (0). Table 37-3 allows implied volatility to fluctuate and shows what the theoretical price of a July 50 call would be, as well as its vega, at those volatilities.

Thus, Table 37-3 shows that vega is surprisingly constant over a wide range of implied volatilities. That’s the real reason why no one bothers with “vega of the vega.” Vega begins to decline only if implied volatility gets exceedingly high, and implied volatilities of that magnitude are relatively rare.

One can also compute the distance a stock would need to rise in order to overcome a decrease in volatility. Consider Figure 37-1, which shows the theoretical price of a 6-month call option with differing implied volatilities. Suppose one buys an option that currently has implied volatility of 170% (the top curve on the graph). Later, investor perceptions of volatility diminish, and the option is trading with an implied volatility of 140%. That means that the option is now “residing” on the *second* curve from the top of the list. Judging from the general distance between those two curves, the option has probably lost between 5 and 8 points of value due to the drop in implied volatility.

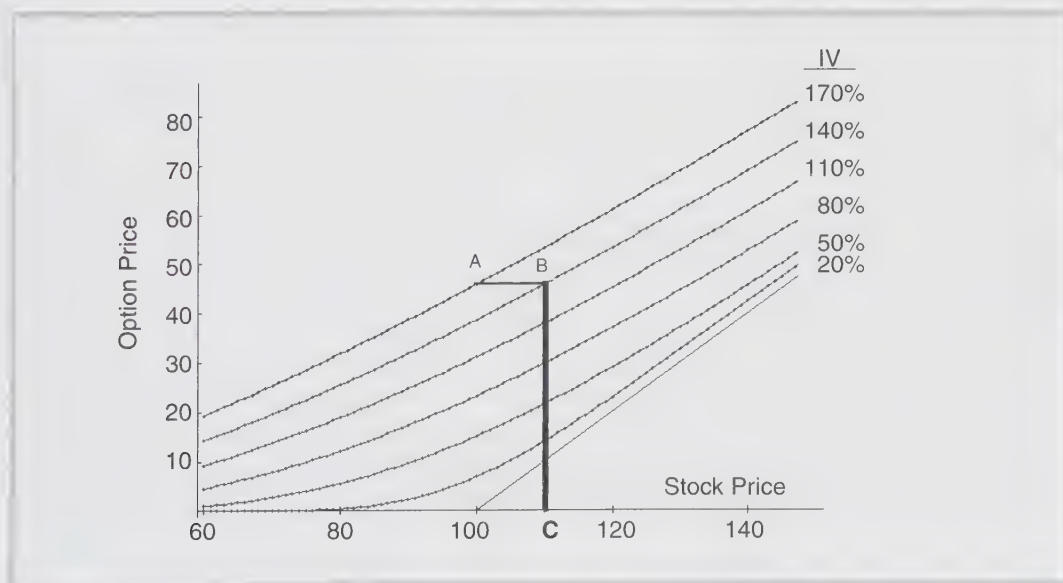
Here’s another way to think about it. Again, suppose one buys an at-the-money option (stock price = 100) when its implied volatility is 170%. That option value is marked as point A on the graph in Figure 37-1. Later, the option’s implied volatility drops to 140%.

TABLE 37-3.

Stock Price	Implied Volatility	Theoretical Call Price	Vega
50	10%	1.34	0.097
	30%	3.31	0.099
	50%	5.28	0.099
	70%	7.25	0.098
	100%	10.16	0.096
	150%	14.90	0.093
	200%	19.41	0.088

FIGURE 37-1.

Theoretical option prices at differing implied volatilities (6-month calls).



How much does the stock have to rise in order to overcome the loss of implied volatility? The horizontal line from point A to point B shows that the option value is the same on each line. Then, dropping a vertical line from B down to point C, we see that point C is at a stock price of about 109. Thus, the stock would have to rise 9 points just to keep the option value constant, *if implied volatility drops from 170% to 140%*.

IMPLIED VOLATILITY AND DELTA

Figure 37-1 shows another rather unusual effect: When implied volatility gets very high, the delta of the option doesn't change much. Simplistically, the delta of an option measures how much the option changes in price when the stock moves one point. Mathematically, the delta is the first partial derivative of the option model with respect to stock price. Geometrically, that means that *the delta of an option is the slope of a line drawn tangent to the curve in the preceding chart.*

The bottom line in Figure 37-1 (where implied volatility = 20%) has a distinct curvature to it when the stock price is between about 80 and 120. Thus the delta ranges from a fairly low number (when the stock is near 80) to a rather high number (when the stock is near 120). Now look at the top line on the chart, where implied volatility = 170%. It's almost a straight line from the lower left to the upper right! The slope of a straight line is constant. This tells us that the delta (which is the slope) *barely changes for such an expensive option—whether the stock is trading at 60 or it's trading at 150!* That fact alone is usually surprising to many.

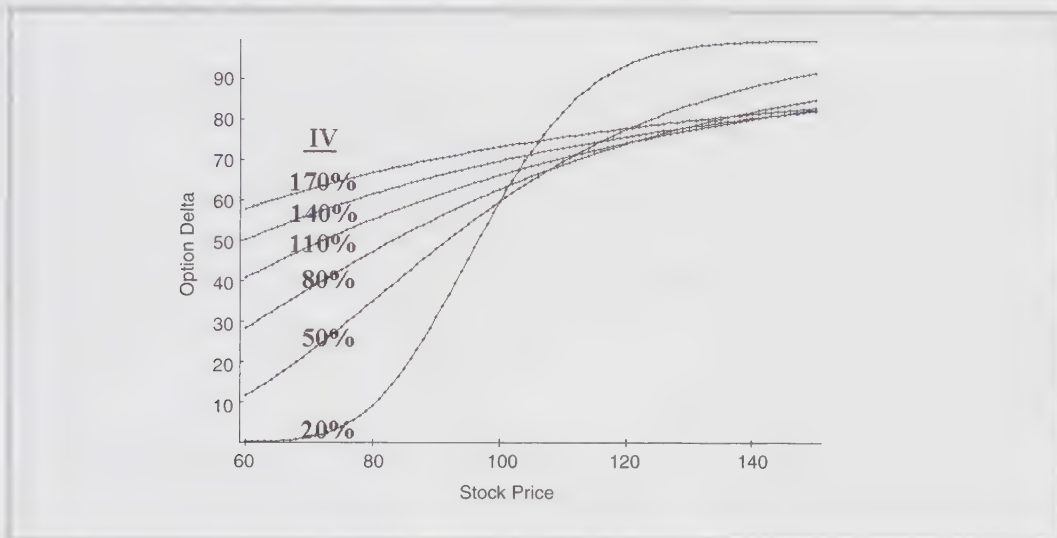
In addition, the value of this delta can be measured: It's 0.70 or higher from a stock price of 80 all the way up to 150. Among other things, this means that an out-of-the-money option that has extremely high implied volatility has a fairly high delta—and can be expected to mirror stock price movements more closely than one might think, were he not privy to the delta.

Figure 37-2 follows through on this concept, showing how the delta of an option varies with implied volatility. From this chart, it is clear how much the delta of an option varies when the implied volatility is 20%, as compared to how little it varies when implied volatility is extremely high.

That data is interesting enough by itself, but it becomes even more thought-provoking when one considers that a change in the implied volatility of his option (vega) also can mean a significant change in the delta of the option. In one sense, it explains why, in the first chart (Figure 37-1), the stock could rise 9 points and yet the option holder made nothing, because implied volatility declined from 170% to 140%.

EFFECTS ON NEUTRALITY

A popular concept that uses delta is the “delta-neutral” spread—a spread whose profitability is supposedly ambivalent to market movement, at least for short time frames and limited stock price changes. Anything that significantly affects the delta of an option can affect this neutrality, thus causing a delta-neutral position to become unbalanced (or,

FIGURE 37-2.**Value of delta of a 6-month option at differing implied volatilities.**

more likely, causing one's intuition to be wrong regarding what constitutes a delta-neutral spread in the first place).

Let's use a familiar strategy, the straddle purchase, as an example. Simplistically, when one buys a straddle, he merely buys a put and a call with the same terms and doesn't get any fancier than that. However, it may be the case that, due to the deltas of the options involved, that approach is biased to the upside, and a neutral straddle position should be established instead.

Example: Suppose that XYZ is trading at 100, that the options have an implied volatility of 40%, and that one is considering buying a six-month straddle with a striking price of 100. The following data summarize the situation, including the option prices and the deltas:

XYZ Common: 100; Implied Volatility: 40%

Option	Price	Delta
XYZ October 100 call	12.00	0.60
XYZ October 100 put	10.00	-0.40

Notice that the stock price is *equal* to the strike price (100). However, the deltas are not at all equal. In fact, the delta of the call is 1.5 times that of the put (in absolute value). One must buy *three puts and two calls* in order to have a delta-neutral position.

Most experienced option traders know that the delta of an at-the-money call is somewhat higher than that of an at-the-money put. Consequently, they often estimate, without checking, that buying three puts and two calls produces a delta-neutral “straddle buy.” However, consider a similar situation, but with a much higher implied volatility—110%, say:

AAA Common: 100; Implied Volatility: 110%

Option	Price	Delta
AAA October 100 call	31.00	0.67
AAA October 100 put	28.00	-0.33

The delta-neutral ratio here is two-to-one (67 divided by 33), not three-to-two as in the earlier case—even though both stock prices are 100 and both sets of options have six months remaining. This is a big difference in the delta-neutral ratio, especially if one is trading a large quantity of options. This shows how different levels of implied volatility can alter one’s perception of what is a neutral position. It also points out that one can’t necessarily rely on his intuition; it is always best to check with a model.

Carrying this thought a step further, one must be mindful of a change in implied volatility if he wants to *keep* his position delta-neutral. If the implied volatility of AAA options should drop significantly, the 2-to-1 ratio will no longer be neutral, even if the stock is still trading at 100. Hence, a trader wishing to remain delta-neutral must monitor not only changes in stock price, but changes in implied volatility as well. For more complex strategies, one will also find the delta-neutral ratio changing due to a change in implied volatility.

The preceding examples summarize the major variables that might affect the vega and also show how vega affects things other than itself, such as delta and, therefore, delta neutrality. By the way, the vega of the underlying is zero; an increase in implied volatility does not affect the price of the underlying instrument at all, in theory. In reality, if options get very expensive (i.e., implied volatility spikes up), that usually brings traders into a stock and so the stock price will change. But that’s not a mathematical relationship, just a market cause-and-effect relationship.

POSITION VEGA

As can be done with delta or with any other of the partial derivatives of the model, one can compute a *position vega*—the vega of an entire position. The position vega is determined

by multiplying the individual option vegas by the quantity of options bought or sold. The “position vega” is merely the quantity of options held, times the vega, times the shares per options (which is normally 100).

Example: Using a simple call spread as an example, assume the following prices exist:

Security	Position	Vega	Position Vega
XYZ Stock	No position		
XYZ July 50 call	Long 3 calls	0.098	+0.294
XYZ July 70 call	Short 5 calls	0.076	−0.380
Net Position Vega:			−0.086

This concept is very important to a volatility trader, for it tells him if he has constructed a position that is going to behave in the manner he expects. For example, suppose that one identifies expensive options, and he figures that implied volatility will decrease, eventually becoming more in line with its historical norms. Then he would want to construct a position with a *negative position vega*. A negative position vega indicates that the position will profit if implied volatility *decreases*. Conversely, a buyer of volatility—one who identifies some underpriced situation—would want to construct a position with a *positive position vega*, for such a position will profit if implied volatility *rises*. In either case, other factors such as delta, time to expiration, and so forth will have an effect on the position’s actual dollar profit, but the concept of position vega is still important to a volatility trader. It does no good to identify cheap options, for example, and then establish some strange spread with a negative position vega. Such a construct would be at odds with one’s intended purpose—in this case, buying cheap options.

OUTRIGHT OPTION PURCHASES AND SALES

Let us now begin to investigate the affects of implied volatility on various strategies, beginning with the simplest strategy of all—the outright option purchase. It was already shown that implied volatility affects the price of an individual call or put in a *direct* manner. That is, an increase in implied volatility will cause the option price to rise, while a decrease in volatility will cause a decline in the option price. That piece of information is the most important one of all, for it imparts what an option trader needs to know: An explosion in implied volatility is a boon to an option owner, but can be a devastating detriment to an option seller, especially a naked option seller.

A couple of examples might demonstrate more clearly just how powerful the effect of

implied volatility is, even when there isn't much time remaining in the life of an option. One should understand the notion that an increase in implied volatility can overcome days, even weeks, of time decay. This first example attempts to quantify that statement somewhat.

Example: Suppose that XYZ is trading at 100 and one is interested in analyzing a 3-month call with striking price of 100. Furthermore, suppose that implied volatility is currently at 20%. Given these assumptions, the Black–Scholes model tells us that the call would be trading at a price of 4.64.

Stock Price:	100
Strike Price:	100
Time Remaining:	3 months
Implied Volatility:	20%
Theoretical Call Value:	4.64

Now, suppose that a month passes. If implied volatility remained the same (20%), the call would lose nearly a point of value due to time decay. However, how much would implied volatility have had to increase to completely counteract the effect of that time decay? That is, after a month has passed, what implied volatility will yield a call price of 4.64? It turns out to be just under 26%.

Stock Price:	100
Strike Price:	100
Time Remaining:	2 months
Implied Volatility:	25.9%
Theoretical Call Value:	4.64

What would happen after *another* month passes? There is, of course, some implied volatility at which the call would still be worth 4.64, but is it so high as to be unreasonable? Actually, it turns out that if implied volatility increases to about 38%, the call will still be worth 4.64, even with only one month of life remaining:

Stock Price:	100
Strike Price:	100
Time Remaining:	1 month
Implied Volatility:	38.1%
Theoretical Call Value:	4.64

So, if implied volatility increases from 20% to 26% over the first month, then this call option would still be trading at the same price—4.64. That's not an unusual increase in implied volatility; increases of that magnitude, 20% to 26%, happen all the time. For it to then increase from 26% to 38% over the *next* month is probably less likely, but it is certainly not out of the question. There have been many times in the past when just such an increase has been possible—during any of the August, September, or October bear markets or mini-crashes, for example. Also, such an increase in implied volatility might occur if there were takeover rumors in this stock, or if the entire market became more volatile, as was the case in the latter half of the 1990s.

Perhaps this example was distorted by the fact that an implied volatility of 20% is a fairly low number to begin with. What would a similar example look like if one started out with a much higher implied volatility—say, 80%?

Example: Making the same assumptions as in the previous example, but now setting the implied volatility to a much higher level of 80%, the Black–Scholes model now says that the call would be worth a price of 16.45:

Stock Price:	100
Strike Price:	100
Time Remaining:	3 months
Implied Volatility:	80%
Theoretical Call Value:	16.45

Again, one must ask the question: “If a month passes, what implied volatility would be necessary for the Black–Scholes model to yield a price of 16.45?” In this case, it turns out to be an implied volatility of just over 99%.

Stock Price:	100
Strike Price:	100
Time Remaining:	2 months
Implied Volatility:	99.4%
Theoretical Call Value:	16.45

Finally, to be able to completely compare this example with the previous one, it is necessary to see what implied volatility would have to rise to in order to offset the effect of yet another month's time decay. It turns out to be over 140%:

Stock Price:	100
Strike Price:	100
Time Remaining:	1 month
Implied Volatility:	140.9%
Theoretical Call Value:	16.45

Table 37-4 summarizes the results of these examples, showing the levels to which implied volatility would have to rise to maintain the call's value as time passes.

Are the volatility increases in the latter example less likely to occur than the ones in the former example? Probably yes—certainly the last one, in which implied volatility would have to increase from 80% to nearly 141% in order to maintain the call's value. However, in another sense, it may seem more reasonable: Note that the increase in volatility from 20% to 26% is a 30% increase. That is, 20% times 1.30 equals 26%. That's what's required to maintain the call's value for the lower volatility over the first month—an increase in the magnitude of implied volatility of 30%. At the *higher* volatility, though, an increase in magnitude of only about 25% is required (from 80% to 99%). Thus, in *those* terms, the two appear on more equal footing.

What makes the top line of Table 37-4 *appear* more likely than the bottom line is merely the fact that an experienced option trader knows that many stocks have implied volatilities that can fluctuate in the 20% to 40% range quite easily. However, there are far fewer stocks that have implied volatilities in the higher range. In fact, until the Internet stocks got hot in the latter portion of the 1990s, the only ones with volatilities like those were very low-priced, extremely volatile stocks. Hence one's experience factor is lower with such high implied volatility stocks, but it doesn't mean that the volatility fluctuations appearing in Table 37-4 are impossible.

If the reader has access to a software program containing the Black–Scholes model, he can experiment with other situations to see how powerful the effect of implied volatility is. For example, without going into as much detail, if one takes the case of a 12-month option whose initial implied volatility is 20%, all it takes to maintain the call's value over

TABLE 37-4.

Initial Implied Volatility	Volatility Level Required to Maintain Call Value . . .	
	. . . After One Month	. . . After Two Months
20%	26%	38%
80%	99%	141%

a 6-month time period is an increase in implied volatility to 27%. Taken from the viewpoint of the option seller, this is perhaps most enlightening: If you sell a one-year (LEAPS) option and six months pass, during which time implied volatility increases from 20% to 27%—certainly quite possible—you will have made nothing! The call will still be selling for the same price, assuming the stock is still selling for the same price.

Finally, it was mentioned earlier that implied volatility often explodes during a market crash. In fact, one could determine just how much of an increase in implied volatility would be necessary in a market crash in order to maintain the call's value. This is similar to the first example in this section, but now the stock price will be allowed to decrease as well. Table 37-5, then, shows what implied volatility would be required to maintain the call's initial value (a price of 4.64), when the stock price falls. The other factors remain the same: time remaining (3 months), striking price (100), and interest rate (5%). Again, this table shows instantaneous price changes. In real life, a slightly higher implied volatility would be necessary, because each market crash could take a day or two.

Thus, from Table 37-5, one could say that even if the underlying stock dropped 20 points (which is 20% in this case) in one day, yet implied volatility exploded from 20% to 67% at the same time, the call's value would be unchanged! Could such an outrageous thing happen? It has: In the Crash of '87, the market plummeted 22% in one day, while the Volatility Index (VIX) theoretically rose from 36% to 150% in one day. In fact, call buyers of some OEX options actually broke even or made a little money due to the explosion in implied volatility, despite the fact that the worst market crash in history had occurred.

If nothing else, these examples should impart to the reader how important it is to be aware of implied volatility at the time an option position is established. If you are buying options, and you buy them when implied volatility is "low," you stand to benefit if implied volatility merely returns to "normal" levels while you hold the position. Of course, having the underlying increase in price is also important.

TABLE 37-5.

Stock Price	Implied Volatility Necessary for Call to Maintain Value
100	20% (the initial parameters)
95	33%
90	44%
85	55%
80	67%
75	78%
70	89%

Conversely, an option seller should be keenly aware of implied volatility when the option is initially sold—perhaps even more so than the buyer of an option. This pertains equally well to naked option writers and to covered option writers. If implied volatility is “too low” when the option writing position is established, then an increase (or worse, an explosion) in implied volatility will be very detrimental to the position, completely overcoming the effects of time decay. Hence, an option writer should not just sell options because he thinks he is collecting time decay each day that passes. That may be true, but an increase in implied volatility can completely dominate what little time decay might exist, especially for a longer-term option.

In a similar manner, a *decrease* in implied volatility can be just as important. Thus, if the call buyer purchases options that are “too costly,” ones in which implied volatility is “too high,” then he could lose money even if the underlying makes a modest move in his favor.

In the next chapters, the topic of just how an option buyer or seller should measure implied volatility to determine what is “too low” or “too high” will be discussed. For now, suffice it to grasp the general concept that a change in implied volatility can have substantial effects on an option’s price—far greater effects than the passage of time can have.

In fact, all of this calls into question just exactly what *time value premium* is. That part of an option’s value that is *not* intrinsic value is really affected much more by volatility than it is by time decay, yet it carries the term “time value premium.”

TIME VALUE PREMIUM IS A MISNOMER

Many (perhaps novice) option traders seem to think of *time* as the main antagonist to an option buyer. However, when one really thinks about it, he should realize that the portion of an option that is *not* intrinsic value is really much more related to stock price movement and/or volatility than anything else, at least in the short term. For this reason, it might be beneficial to more closely analyze just what the “excess value” portion of an option represents and why a buyer should not primarily think of it as time value premium.

An option’s price is composed of two parts: (1) intrinsic value, which is the “real” part of the option’s value—the distance by which the option is in-the-money, and (2) “excess value”—often called time value premium. There are actually five factors that affect the “excess value” portion of an option. Eventually, time will dominate them all, but the longer the life of the option, the more the other factors influence the “excess value.”

The five factors influencing excess value are:

1. stock price movements,
2. changes in implied volatility,
3. the passage of time,

4. changes in the dividend (if any exist), and
5. changes in interest rates.

Each is stated in terms of a movement or change; that is, these are not static things. In fact, to measure them one uses the “greeks”: delta, vega, theta (there is no “greek” for dividend change), and rho. Typically, the effect of a change in dividend or a change in interest rate is small (although a large dividend change or an interest rate change on a very long-term option can produce visible changes in the prices of options).

If everything remains static, then time decay will eventually wipe out all of the excess value of an option. That’s why it’s called time value premium. But things don’t ever remain static, and on a daily basis, time decay is small, so it is the remaining two factors that are most important.

Example: XYZ is trading at 82 in late November. The January 80 call is trading at 8. Thus, the intrinsic value is 2 (82 minus 80) and the excess value is 6 (8 minus 2). If the stock is still at 82 at January expiration, the option will of course only be worth 2, and one will say that the 6 points of excess value that was lost was due to time decay. But on that day in late November, the other factors are much more dominant.

On this particular day, the implied volatility of this option is just over 50%. One can determine that the call’s greeks are:

Delta: 0.60

Vega: 0.13

Theta: -0.06

This means, for example, that time decay is only 6 cents per day. It would increase as time went by, but even with a day or so to go, theta would not increase above about 20 cents unless volatility increased or the stock moved closer to the strike price.

From the above figures, one can see—and this should be intuitively appealing—that the biggest factor influencing the price of the option is stock price movement (delta). It’s a little unfair to say that, because it’s conceivable (although unlikely) that volatility could jump by a large enough margin to become a greater factor than delta for one day’s move in the option. Furthermore, since this option is composed mostly of excess value, these more dominant forces influence the excess value more than time decay does.

There is a direct relationship between vega and excess value. That is, if implied volatility increases, the excess value portion of the option will increase and, if implied volatility decreases, so will excess value.

The relationship between delta and excess value is not so straightforward. The farther the stock moves away from the strike, the more this will have the effect of shrinking the excess value. If the call is in-the-money (as in the above example), then an *increase* in stock price will result in a *decrease* of excess value. That is, a deeply in-the-money option is composed primarily of intrinsic value, while excess value is quite small. However, when the call is out-of-the-money, the effect is just the opposite: Then, an increase in call price will result in an increase in excess value, because the stock price increase is bringing the stock closer to the option's striking price.

For some readers, the following may help to conceptualize this concept. The part of the delta that addresses excess value is this:

Out-of-the-money call: 100% of the delta affects the excess value.

In-the-money call: "1.00 minus delta" affects the excess value. (So, if a call is very deeply in-the-money and has a delta of 0.95, then the delta only has 1.00—0.95, or 0.05, room to increase. Hence it has little effect on what small amount of excess value remains in this deeply in-the-money call.)

These relationships are not static, of course. Suppose, for example, that in the same situation of the stock trading at 82 and the January 80 call trading at 8, *there is only one week remaining until expiration!* Then the implied volatility would be 155% (high, but not unheard of in volatile times). The greeks would bear a significantly different relationship to each other in this case, though:

Delta: 0.59

Vega: 0.044

Theta: -0.51

This very short-term option has about the same delta as its counterpart in the previous example (the delta of an at-the-money option is generally slightly above 0.50). Meanwhile, vega has shrunk. The effect of a change in volatility on such a short-term option is actually about a third of what it was in the previous example. However, time decay in this example is huge, amounting to half a point per day in this option.

So now one has the idea of how the excess value is affected by the "big three" of stock price movement, change in implied volatility, and passage of time. How can one use this to his advantage? First of all, one can see that an option's excess value may be due much more to the potential volatility of the underlying stock, and therefore to the option's implied volatility, than to time.

As a result of the above information regarding excess value, one shouldn't think that he can easily go around selling what appear to be options with a lot of excess value and then

expect time to bring in the profits for him. In fact, there may be a lot of volatility—both actual and implied—keeping that excess value nearly intact for a fairly long period of time. In fact, in the coming chapters on volatility estimation, it will be shown that option buyers have a much better chance of success than conventional wisdom has maintained.

VOLATILITY AND THE PUT OPTION

While it is obvious that an increase in implied volatility will increase the price of a put option, much as was shown for a call option in the preceding discussion, there are certain differences between a put and a call, so a little review of the put option itself may be useful. A put option tends to lose its premium fairly quickly as it becomes an in-the-money option. This is due to the realities of conversion arbitrage. In a conversion arbitrage, an arbitrageur or market-maker buys stock *and* buys the put, while selling the call. If he carries the position to expiration, he will have to pay carrying costs on the debit incurred to establish the position. Furthermore, he would earn any dividends that might be paid while he holds the position. This information was presented in a slightly different form in the chapter on arbitrage, but it is recounted here:

In a perfect world, all option prices would be so accurate that there would be no profit available from a conversion. That is, the following equation (1) would apply:

$$(1) \text{ Call price} + \text{Strike price} - \text{Stock price} - \text{Put price} + \text{Dividend} - \text{Carrying cost} = 0$$

$$\text{where carrying cost} = \text{strike price} / (1 + r)^t$$

$$t = \text{time to expiration}$$

$$r = \text{interest rate}$$

Now, it is also known that the time value premium of a put is the amount by which its value exceeds intrinsic value. The intrinsic value of an in-the-money put option is merely the difference between the strike price and the stock price. Hence, one can write the following equation (2) for the time value premium (TVP) of an in-the-money put option:

$$(2) \text{ Put TVP} = \text{Put price} - \text{Strike price} + \text{Stock price}$$

The arbitrage equation, (1), can be rewritten as:

$$(3) \text{ Put price} - \text{Strike price} + \text{Stock price} = \text{Call price} + \text{Dividends} - \text{Carrying cost}$$

and substituting equation (2) for the terms in equation (3), one arrives at:

$$(4) \text{ Put TVP} = \text{Call price} + \text{Dividends} - \text{Carrying cost}$$

In other words, the time value premium of an in-the-money put is the same as the (out-of-the-money) call price, plus any dividends to be earned until expiration, less any carrying costs over that same time period.

Assuming that the dividend is small or zero (as it is for most stocks), one can see that an in-the-money put would lose its time value premium whenever carrying costs exceed the value of the out-of-the-money call. Since these carrying costs can be relatively large (the carrying cost is the interest being paid on the entire debit of the position—and that debit is approximately equal to the strike price), they can quickly dominate the price of an out-of-the-money call. Hence, the time value premium of an in-the-money put disappears rather quickly.

This is important information for put option buyers, because they must understand that a put won't appreciate in value as much as one might expect, even when the stock drops, since the put loses its time value premium quickly. It's even more important information for put *sellers*: A short put is at risk of assignment as soon as there is no time value premium left in the put. Thus, a put can be assigned well in advance of expiration—even a LEAPS put!

Now, returning to the main topic of how implied volatility affects a position, one can ask himself how an increase or decrease in implied volatility would affect equation (4) above. If implied volatility increases, the call price would increase, and if the increase were great enough, might impart some time value premium to the put. *Hence, an increase in implied volatility also may increase the price of a put, but if the put is too far in-the-money, a modest increase in implied volatility still won't budge the put.* That is, an increase in implied volatility would increase the value of the call, but until it increases enough to be greater than the carrying costs, an in-the-money put will remain at parity, and thus a short put would still remain at risk of assignment.

STRADDLE OR STRANGLE BUYING AND SELLING

Since owning a straddle involves owning both a put and a call with the same terms, it is fairly evident that an increase in implied volatility will be very beneficial for a straddle buyer. A sort of double benefit occurs if implied volatility rises, for it will positively affect both the put and the call in a long straddle. Thus, if a straddle buyer is careful to buy straddles in situations in which implied volatility is “low,” he can make money in one of two ways. Either (1) the underlying price makes a move great enough in magnitude to exceed the initial cost of the straddle, or (2) implied volatility increases quickly enough to overcome the deleterious effects of time decay.

Conversely, a straddle seller risks just the opposite—potentially devastating losses if implied volatility should increase dramatically. However, the straddle seller can register

gains faster than just the rate of time decay would indicate if implied volatility *decreases*. Thus, it is very important when selling options—and this applies to covered options as well as to naked ones—to sell only when implied volatility is “high.”

A strangle is the same as a straddle, except that the call and put have different striking prices. Typically, the call strike price is higher than the put strike price. Naked option sellers often prefer selling strangles in which the options are well out-of-the-money, so that there is less chance of them having any intrinsic value when they expire. Strangles behave much like straddles do with respect to changes in implied volatility.

The concepts of straddle ownership will be discussed in much more detail in the following chapters. Moreover, the general concept of option buying versus option selling will receive a great deal of attention.

CALL BULL SPREADS

In this section, the bull spread strategy will be examined to see how it is affected by changes in implied volatility. Let's look at a call bull spread and see how implied volatility changes might affect the price of the spread if all else remains equal. Make the following assumptions:

Assumption Set 1:

Stock Price: 100

Time to Expiration: 4 months

Position: Long Call Struck at 90

Short Call Struck at 110

Ask yourself this simple question: If the stock remains unchanged at 100, and implied volatility increases dramatically, will the price of the 90–110 call bull spread grow or shrink? Answer before reading on.

The truth is that, if implied volatility *increases*, the price of the spread will *shrink*. I would suspect that this comes as something of a surprise to a good number of readers. Table 37-6 contains some examples, generated from a Black–Scholes model, using the assumptions stated above, the most important of which is that the stock is at 100 in all cases in this table.

One should be aware that it would probably be difficult to actually trade the spread at the theoretical value, due to the bid-asked spread in the options. Nevertheless, the impact of implied volatility is clear.

One can quantify the amount by which an option position will change for each per-

TABLE 37-6.

Stock Price = 100	
Implied Volatility	90–110 Call Bull Spread (Theoretical Value)
20%	10.54
30%	9.97
40%	9.54
50%	9.18
60%	8.87
70%	8.58
80%	8.30

centage point of increase in implied volatility. Recall that this measure is called the vega of the option or option position. In a call bull spread, one would subtract the vega of the call that is sold from that of the call that is bought in order to arrive at the position vega of the call bull spread. Table 37-7 is a reprint of Table 37-6, but now including the vega.

Since these vegas are all negative, they indicate that the spread will shrink in value if implied volatility rises and that the spread will expand in value if implied volatility decreases. Again, these statements may seem contrary to what one would expect from a bullish call position.

Of course, it's highly unlikely that implied volatility would change much in the course of just one day while the stock price remained unchanged. So, to get a better handle on what to expect, one really needs to look at what might happen at some future

TABLE 37-7.

Implied Volatility	90–110 Call Bull Spread (Theoretical Value)	Position Vega
20%	10.54	–0.67
30%	9.97	–0.48
40%	9.54	–0.38
50%	9.18	–0.33
60%	8.87	–0.30
70%	8.58	–0.28
80%	8.30	–0.26

time (say a couple of weeks hence) at various stock prices. The graph in Figure 37-3 begins the investigation of these more complex scenarios.

The profit curve shown in Figure 37-3 makes certain assumptions: (1) The bull spread assumes the details in Assumption Set 1, above; (2) the spread was bought with an implied volatility of 20% and remained at that level when the profit picture above was drawn; and (3) 30 days have passed since the spread was bought. Under these assumptions, the profit graph shows that the bull spread conforms quite well to what one would expect; that is, the shape of this curve is pretty much like that of a bull spread at expiration, although if you look closely you'll see that it doesn't widen out to nearly its maximum gain or loss potential until the stock is well above 110 or below 90—the strike prices used in the spread.

Now observe what happens if one keeps all the other assumptions the same, except one. In this case, assume implied volatility was 80% at purchase and remains at 80% one month later. The comparison is shown in Figure 37-4. The 80% curve is overlaid on top of the 20% curve shown earlier. The contrast is quite startling. Instead of looking like a bull spread, the profit curve that uses 80% implied volatility is a rather flat thing, sloping only slightly upward—and exhibiting far less risk and reward potential than its lower implied volatility counterpart. This points out another important fact: *For volatile stocks, one cannot expect a 4-month bull spread to expand or contract much during the first month of life, even if the stock makes a substantial move.* Longer-term spreads have even less movement.

FIGURE 37-3.
Bull spread profit picture in 30 days, at 20% IV.

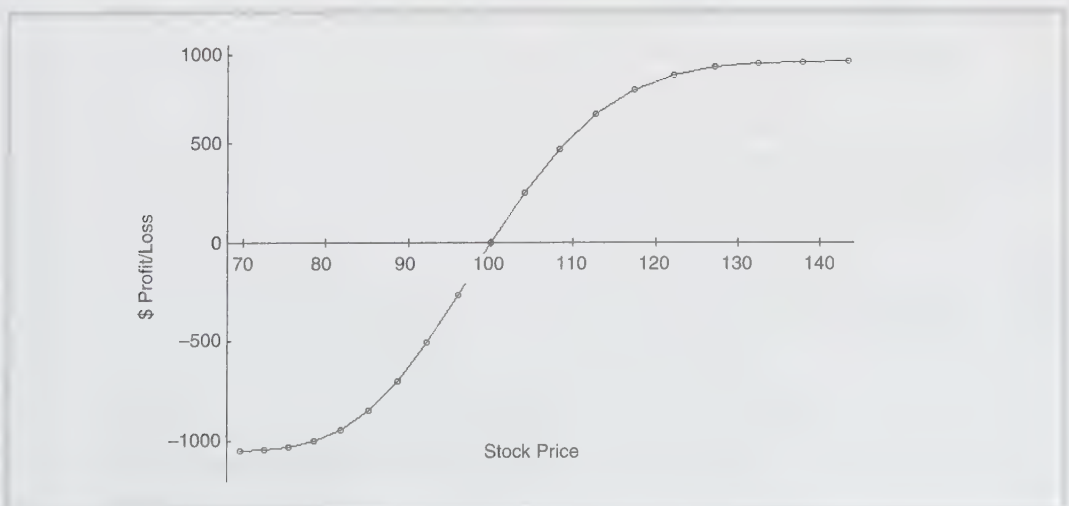
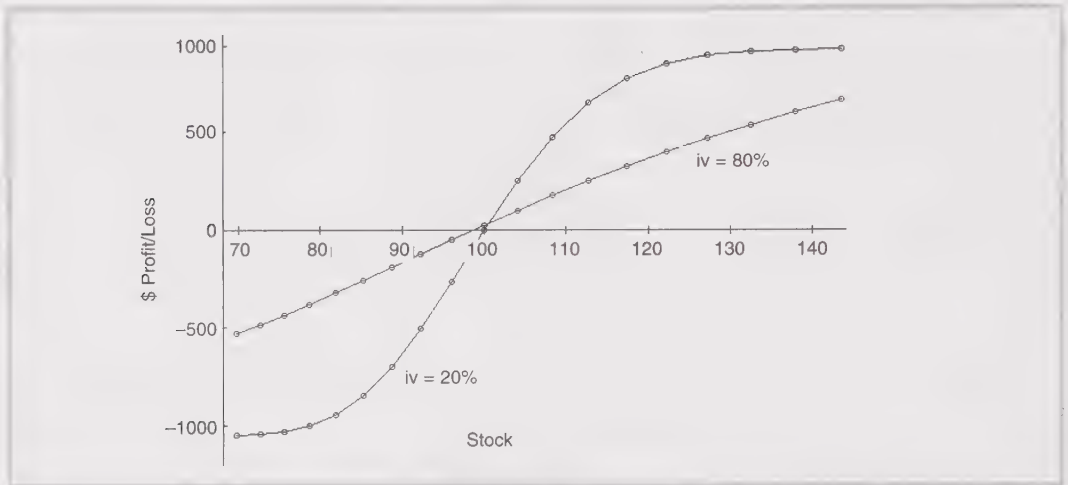


FIGURE 37-4.
Bull spread profit picture in 30 days.



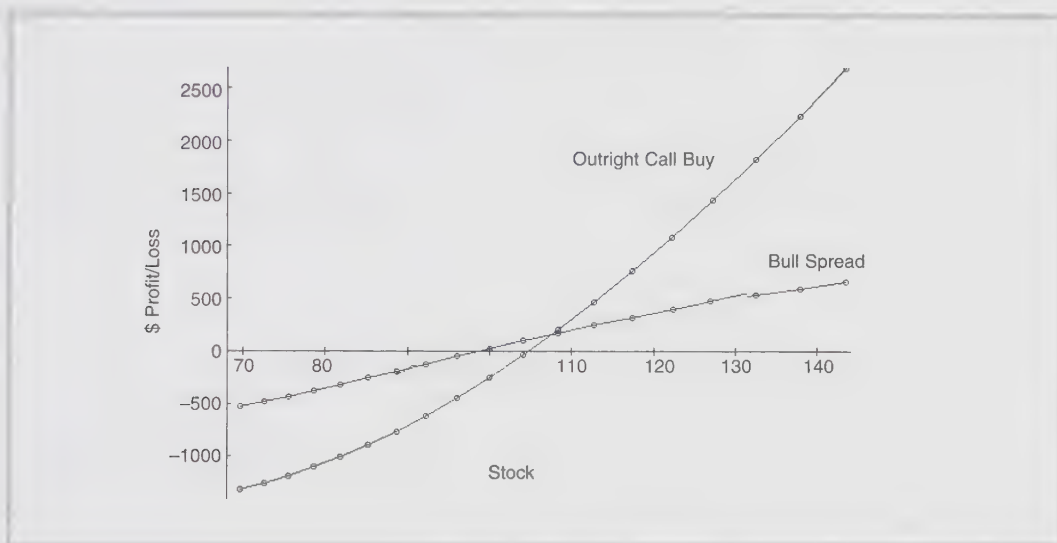
As a corollary, note that if implied volatility *shrinks* while the stock rises, the profit outlook will improve. Graphically, using Figure 37-4, if one's profit picture moves from the 80% curve to the 20% curve on the right-hand side of the chart, that is a positive development. Of course, if the stock drops and the implied volatility drops too, then one's losses would be worse—witness the left-hand side of the graph in Figure 37-4.

A graph could be drawn that would incorporate other implied volatilities, but that would be overkill. The profit graphs of the other spreads from Tables 37-6 or 37-7 would lie between the two curves shown in Figure 37-4.

If this discussion had looked at bull spreads as *put credit* spreads instead of call debit spreads, perhaps these conclusions would not have seemed so unusual. Experienced option traders already understand much of what has been shown here, but less experienced traders may find this information to be different from what they expected.

Some general facts can be drawn about the bull spread strategy. Perhaps the most important one is that, if used on a volatile stock, you won't get much expansion in the spread even if the stock makes a nice move upward in your favor. In fact, for high implied volatility situations, the bull spread won't expand out to its maximum price until expiration draws nigh. That can be frustrating and disappointing.

Often, the bull spread is established because the option trader feels the options are "too expensive" and thus the spread strategy is a way to cut down on the total debit invested. However, the ultimate penalty paid is great. Consider the fact that, if the stock

FIGURE 37-5.**Call buy versus bull spread in 30 days; IV = 80%.**

rose from 100 to 130 in 30 days, *any* reasonable four-month call purchase (i.e., with a strike initially near the current stock price) would make a nice profit, while the bull spread barely ekes out a 5-point gain. To wit, the graph in Figure 37-5 compares the purchase of the at-the-money call with a striking price of 100 and the 90–110 call bull spread, both having implied volatility of 80%. Quite clearly, the call purchase dominates to a great extent on an upward move. Of course, the call purchase does worse on the downside, but since these are bullish strategies, one would have to assume that the trader had a positive outlook for the stock when the position was established. Hence, what happens on the downside is not primary in his thinking.

The bull spread and the call purchase have opposite position vegas, too. That is, a rise in implied volatility will help the call purchase but will harm the bull spread (and vice versa). *Thus, the call purchase and the bull spread are not very similar positions at all.*

If one wants to use the bull spread to effectively reduce the cost of buying an expensive at-the-money option, then at least make sure the striking prices are quite wide apart. That will allow for a reasonable amount of price appreciation in the bull spread if the underlying rises in price. Also, one might want to consider establishing the bull spread with striking prices that are *both* out-of-the-money. Then, if the stock rallies strongly, a greater percentage gain can be had by the spreader. Still, though, the facts described above cannot be overcome; they can only possibly be mitigated by such actions.

A FAMILIAR SCENARIO?

Often, one may be deluded into thinking that the two positions are more similar than they are. For example, one does some sort of analysis—it does not matter if it's fundamental or technical—and comes to a conclusion that the stock (or futures contract or index) is ready for a bullish move. Furthermore, he wants to use options to implement his strategy. But, upon inspecting the actual market prices, he finds that the options seem rather expensive. So, he thinks, "Why not use a bull spread instead? It costs less and it's bullish, too."

Fairly quickly, the underlying moves higher—a good prediction by the trader, and a timely one as well. If the move is a violent one, especially in the futures market, implied volatility might increase as well. If you had bought calls, you'd be a happy camper. But if you bought the bull spread, you are not only highly disappointed, but you are now facing the prospect of having to hold the spread for several more weeks (perhaps months) before your spread widens out to anything even approaching the maximum profit potential.

Sound familiar? Every option trader has probably done himself in with this line of thinking at one time or another. At least, now you know the reason why: High or increasing implied volatility is not a friend of the bull spread, while it is a friendly ally of the outright call purchase. Somewhat surprisingly, many option traders don't realize the difference between these two strategies, which they probably consider to be somewhat similar in nature.

So, be careful when using bull spreads. If you really think a call option is too expensive and want to reduce its cost, try this strategy: Buy the call and simultaneously sell a credit put spread (bull spread) using slightly out-of-the-money puts. This strategy reduces the call's net cost and maintains upside potential (although it increases downside risk, but at least it is still a fixed risk).

Example: With XYZ at 100, a trader is bullish and wants to buy the July 100 calls, which expire in two months. However, upon inspection, he finds that they are trading at 10—an implied volatility of 59%. He knows that, historically, the implied volatility of this stock's options range from approximately 40% to 60%, so these are very expensive options. If he buys them now and implied volatility returns to its median range near 50%, he will suffer from the decrease in implied volatility.

As a possible remedy, he considers selling an out-of-the-money put credit spread at the same time that he buys the calls. The credit from this spread will act as a means of reducing the net cost of the calls. If he's right and the stock goes up, all will be well. However, the introduction of the put spread into the mix has introduced some additional downside risk.

Suppose the following prices exist:

XYZ: 100
 July 100 call: 10 (as stated above)
 July 90 put: 5
 July 80 put: 2

The entire bullish position would now consist of the following:

Buy 1 July 100 call at 10
 Buy 1 July 80 put at 2
 Sell 1 July 90 put at 5
 Net expenditure: 7 point debit (plus commission)

Figure 37-6 shows the profitability, at expiration, of both the outright call purchase and the bullish position constructed above.

First, one can see that the bullish spread position has a total risk of 17 points, if XYZ is below 80 (the lower striking price of the put spread) at expiration. That, of course, is more than the 10-point cost of the July 100 call by itself, but if one is using a trading stop of any sort, he probably would not be at risk for the entire 17 points, since he wouldn't

FIGURE 37-6.
Profitability at expiration.

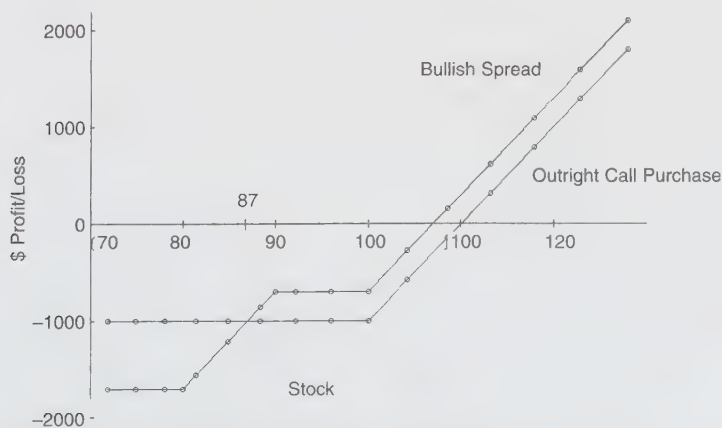
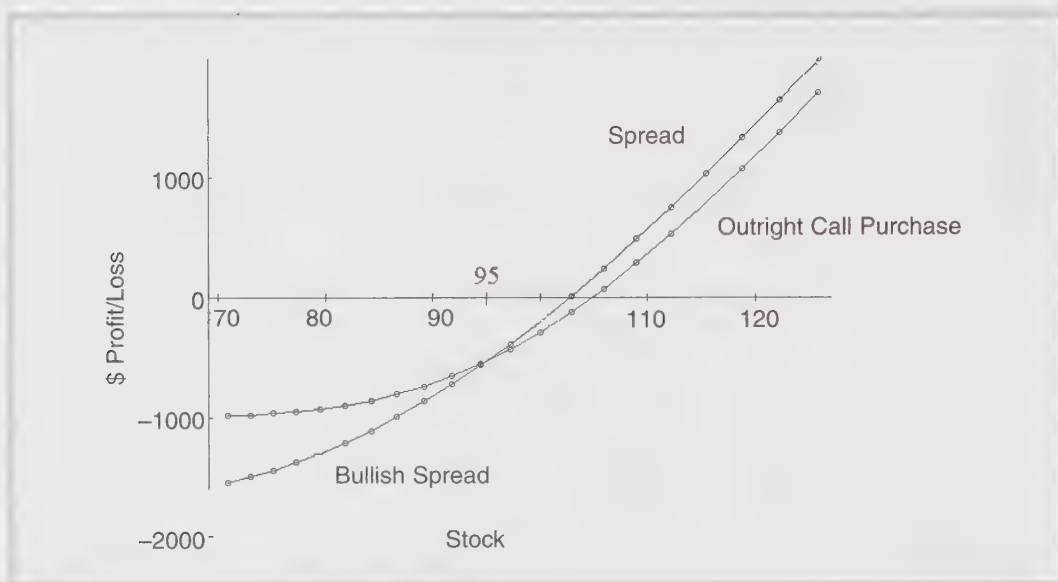


FIGURE 37-7.
Results of the two positions in 30 days.



hold on while the stock fell all the way to 80 and below. Note also that the bullish spread position would have a loss of 10 points (the same as the call) at a price of 87 for the common at expiration. Hence, the combined position actually has *less* risk than the outright call purchase as long as XYZ is 87 or higher at expiration. Since one is supposedly bullish initially when establishing this strategy, it seems likely that he would figure the stock would be 87 or higher at expiration.

Figure 37-7 offers another comparison, that of the two positions after 30 days have passed. Note that the spread position once again does better on the upside and worse on the downside. The crossover point between the two curves is at about a price of 95. That is, if XYZ is above 95 in 30 days, the bullish spread position will outperform the call buy.

One final point should be made regarding the investment required. The outright call purchase requires an investment of \$1,000—the cost of the long call. The bullish spread position requires that \$1,000, plus \$700 for the spread (10-point difference in the strikes, less the 3-point credit received for selling the spread). That's a total of \$1,700, the risk of the bullish spread position. Hence, the *rate of return* might favor the outright call purchase, depending on how far the stock rallies.

Overall, the bullish spread position is an attractive alternative to an outright call purchase, especially when the call is overpriced. The spread does risk a greater amount of money if the underlying stock should collapse heavily. Still, if one is truly bullish, and

if one employs a reasonably tight downside stop on his entire position, this spread can perform better than the outright purchase of an overpriced call.

VERTICAL PUT SPREADS

Also of interest is the effect that implied volatility has on put spreads. One of the more popular strategies involving puts is the sale of a credit spread—a bull spread with puts. Assume that a stock is selling at 100, and one is going to *sell a put with a 110 strike and buy a put with a 90 strike*. That is a put credit (bull) spread. Also assume that the options have four months of life remaining. (See Table 37-8.)

One would not rationally sell this credit spread if implied volatility were as low as 20%, because at that low level of volatility, the in-the-money December 110 put is trading for 10 dollars—parity—and thus would immediately be at risk of early assignment. But one can see that an increase in implied volatility increases the value of the spread. Now, if one had sold this spread to begin with, he would thus be *losing* money when implied volatility increased. This was proven with call bull spreads, too: They lose money when implied volatility increases. Conversely, of course, the put credit spread *makes* money when implied volatility decreases.

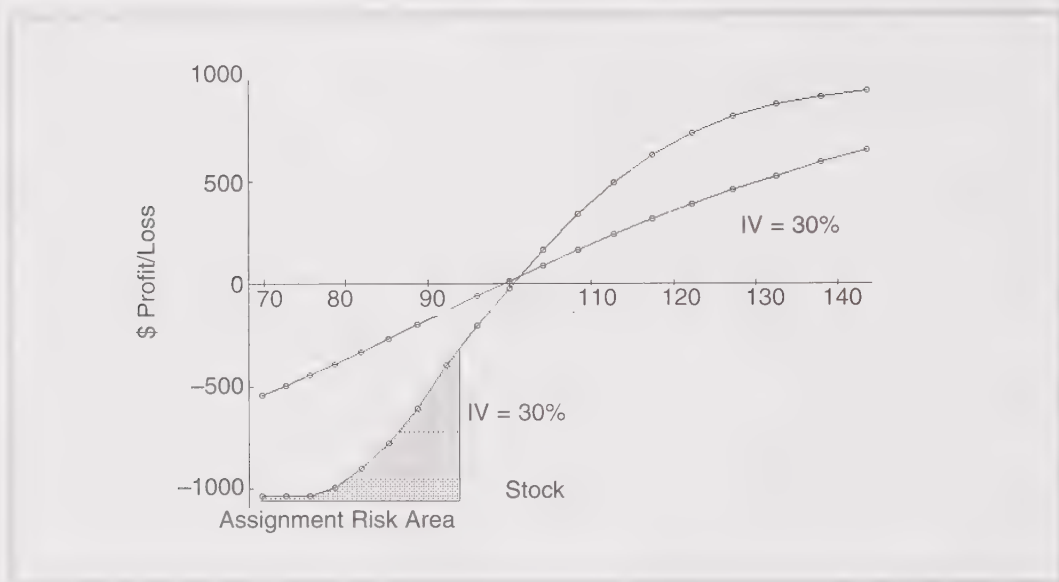
What happens after thirty days have passed? Figure 37-8 shows just two cases—implied volatility at 30% and implied volatility at 80%. One can surmise that other levels of implied volatility between 30% and 80% would have profit graphs that lie between the two shown in Figure 37-8.

First, one can observe that a bull put spread does not widen out to anywhere near its maximum potential if implied volatility increases. The same thing was seen with the call

TABLE 37-8.

Implied Volatility	90–110 Put Bull Spread (Theoretical Value)
20%	9.15 cr*
30%	9.70 cr
40%	10.12 cr
50%	10.46 cr
60%	10.78 cr
70%	11.05 cr
80%	11.33 cr

*Short option trading at parity

FIGURE 37-8.**Put credit (bull) spread profit in 30 days.**

bull spread in the previous section. But a put bull spreader is caught in another trap: If implied volatility falls and the stock falls too, the risk of early assignment materializes quickly. Note the shaded area at the lower left of the graph, extending from a price of about 94 on down. After thirty days (so there would be three months' life remaining at that point in time), if implied volatility is 30%, the 110 put (the short put) would be trading at parity for stock prices of 94 and below. Thus, it would be at risk of early assignment. If implied volatility were even lower, the puts would be at parity for much higher stock prices.

Now, in and of itself, early assignment on an equity or futures put spread is not necessarily a terrible thing. There will be a request for additional margin (because the stock has to be paid for or the futures contract margined), but the risk is still the same in dollar terms. Of course, the request for extra margin could be backbreaking for a stock trader if he can't afford to fully pay for the stock, and the early assignment would probably incur additional commission costs, too. However, with cash-based index options there is a more serious increase in risk after an early assignment, because one is left with only the long side of the spread. If that option happens to have substantial value, then there is considerable risk if the underlying should quickly move higher. In fact, by the time one unwinds the spread, he might actually end up losing more than his original limited risk amount—all due to the early assignment. (This could happen if the underlying first plunges in price, placing both options deeply in-the-money, after which one gets assigned on the short put option, followed by the underlying then dramatically rising in price.)

The lesson to be learned is this: *If one is considering using bull spreads in which at least one of the options is at- or in-the-money, then a call bull spread is a superior choice over a put bull spread.* Early assignment is not really a consideration for most call spreads.

In both cases, however, a more serious problem exists, and that is that the spread does not widen out much even when the stock makes a nice bullish move. Thus, once again it is actually better to buy a call option in most cases than to use the bull spread, because profits are larger and an increase in implied volatility is a favorable thing for an outright call buyer.

Note that these effects are similar, but much less pronounced, for out-of-the-money put credit spreads. Still, it should be noted that an increase in implied volatility will harm an out-of-the-money put credit spread, too. Hence, if the underlying goes into a rapid fall (crash, plunge), then implied volatility usually increases quickly and dramatically. So an out-of-the-money credit spreader is hit with the double whammy of expanding implied volatility and the fact that the underlying is fast approaching the strike price of his options, thereby expanding the price of the spread.

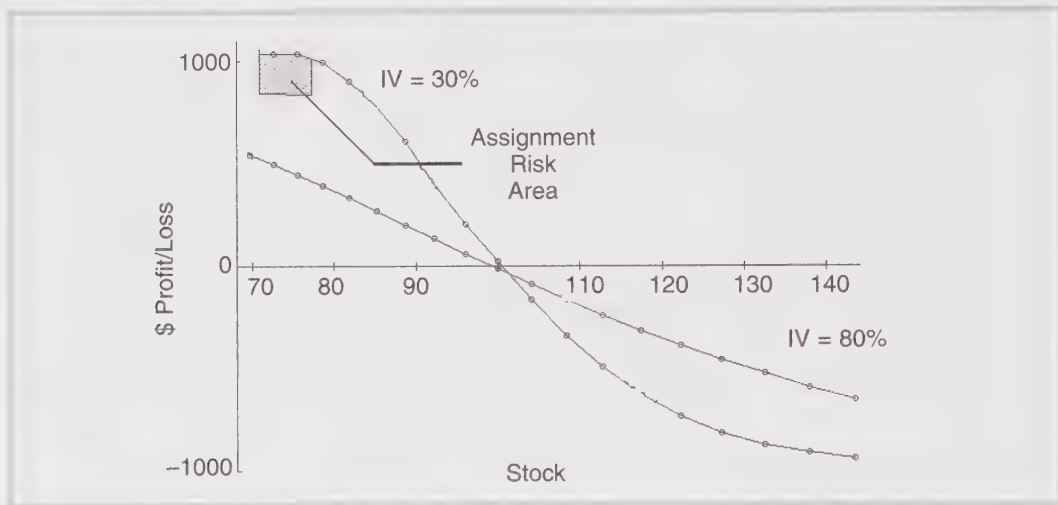
PUT BEAR SPREADS

What about the put spread in a bearish situation? In a vertical put spread one *buys* the put with the higher strike and *sells* the put with the lower strike to construct a simple put bear spread. Actually, a sudden increase in implied volatility is of help to the bear put spread. That is, the spread will widen out slightly. To verify this, look at Table 37-8 again, only now imagine that one is *buying* the spread for a debit. Note that the smallest debit is at the lower implied volatility—9.15 debit with IV at 30%. If implied volatility were to instantaneously jump to 80%, the spread would widen out to 11.33 debit. A very quick profit could be had. So there's a difference right away between a debit call bull spread (which *loses* money when implied volatility suddenly increases) and a debit put bear spread.

Unfortunately, the other major drawback—that the spread doesn't widen out much if the underlying makes a favorable move—is still true. Figure 37-9 shows a bear put spread, 30 days hence, for two different implied volatilities. Once again, the lower-volatility spread widens out more quickly, because both options tend to go to parity in that case. In fact, one can see on the graph that there is early assignment risk in the low-volatility case, below a price of about 77 on the stock. That is not a problem, though, since the spread would have widened to its maximum potential in that case and could just be removed when the risk of early assignment materialized.

When implied volatility remains high, though, the spread doesn't widen out much, even when the stock drops a lot after 30 days. Since it is common for implied volatility to rise (even skyrocket) when the underlying drops quickly, the put bear spread probably

FIGURE 37-9.
Bear put spread profit in 30 days.



won't widen out much. That may not be a psychologically pleasing strategy, because one won't make the level of profits that he had hoped to when the underlying fell quickly.

Once again, it seems that the outright purchase of an option is probably superior to a spread. In these cases, it is true with respect to puts, much as it was with call options. Spreading often unnecessarily complicates a trader's outlook.

CALENDAR SPREADS

In the earlier chapter on calendar spreads, it was mentioned that an increase in implied volatility will cause a calendar spread to widen out. Both options will become more expensive, of course, since the increase in implied volatility affects both of them, but the *absolute* price change will be greatest in the long-term option. Therefore, the calendar spread will widen. This may seem somewhat counterintuitive, especially where highly volatile stocks are concerned, so some examples may prove useful.

Example: Suppose that XYZ is trading at 100, and one is interested in a calendar spread in which an August (5-month) call is bought and a May (2-month) call is sold. For the purpose of this example, it will be assumed that these are both at-the-money options. First, the vegas of the two options will be examined, assuming that implied volatility is 40%:

Stock: 100

Implied Volatility:

40% Option	Theoretical Price	Vega
Sell May 100 call	6.91	0.162
Buy August 100 call	11.22	0.251

In theory, this spread should be worth 4.31—the difference in the theoretical values. Perhaps more important, it has volatility exposure of 0.089—the difference between the vega of the long call and that of the short call. Since vega is positive, this means that an *increase* in implied volatility will be beneficial to the spread. In other words, one can expect the spread to widen if implied volatility rises, and can expect the spread to shrink if implied volatility declines.

The following table can also be constructed, showing the theoretical value of the spread at various levels of implied volatility. This table makes the assumption that very little time has passed (only one week) before the implied volatility changes take place. It also assumes that the stock is still at 100.

Stock Price: 100

One week after the spread has been established:

Implied Volatility	Theoretical Spread Value
20%	2.58
30%	3.52
40%	4.46
50%	5.40
60%	6.33
80%	8.16
100%	12.92

From the above data, it is quite obvious that implied volatility levels have a huge effect on the value of a calendar spread. The actual initial contribution of time decay is rather small in comparison. For example, note that if volatility remains unchanged at 40%, then the spread will have widened only slightly—to 4.46 from 4.31—after the passage of one week's time. That is small in comparison to the changes dictated by volatility expansion or contraction.

A common mistake that calendar spreaders make is to think that such a spread looks overly attractive on a very volatile stock. Consider the same stock as above, still trading at

100, but for some reason implied volatility has skyrocketed to 80% (perhaps a takeover rumor is present).

Stock: 100	
Implied Volatility: 80%	
Call	Theoretical Value
May 100 call	12.55
June 100 call	16.81

On the surface, this seems like a very attractive spread. There are two months of life remaining in the May options (and three months in the Junes) and the spread is trading at 4.36. However, both options are completely composed of time value premium, and most certainly the June 100 call would be worth far more than 4.36 when the May expires, if the stock is still near 100. The fact that many traders miss when they think of the calendar spread this way is that the June call will only be worth “far more than 4.36” if implied volatility holds up. If implied volatility for this stock is normally something on the order of 40%, say, then it is probably not reasonable to expect that the 80% level will hold up. Just for comparison, note that if the stock is at 100 at May expiration—the maximum profit potential for such a calendar spread—the June 100 call, with implied volatility now at 40%, and with one month of life remaining, would be worth only 4.77. Thus the spread would only have made a profit of a few cents (4.36 to 4.77), and if the underlying stock were farther from the strike price at expiration, there would probably be a loss rather than a profit.

The point to be remembered is that a calendar spread is a “long volatility” play (and a reverse calendar spread is just the opposite). Evaluate the position’s risk with an eye to what might happen to implied volatility, and not just to where the stock price might go or how much time decay there might be in the position.

RATIO SPREADS AND BACKSPREADS

The previous descriptions in this chapter describe fairly fully and accurately what the effect of volatility changes are. More complicated strategies are usually nothing more than combinations of the strategies presented earlier, so it is easy to discern the effect that changes in implied volatility would have; just combine the effects on the simpler strategies. For example, a ratio call write is really just the equivalent of a straddle sale—a strategy whose volatility ramifications are fairly simple to understand.

Ratio spreads, on the other hand, might not be as intuitive to interpret, but they are fairly simple nonetheless. A call ratio spread is really just the combination of some call bull

spreads and some naked call options. For example, a call ratio spread might consist of buying an XYZ July 100 call and selling two XYZ July 120 calls. If one were to break it down into its components, this spread is really long one XYZ July 100–120 call bull spread, plus an additional naked July 120 call.

We already know that an increase in implied volatility is very detrimental to a naked call option. In addition, it was shown earlier than an increase in implied volatility actually *harms* the value of an at-the-money call bull spread. So, for a ratio call spread, *both* components are harmed by an increase in implied volatility. Conversely, a *decrease* in implied volatility would be beneficial to a ratio spread, but where naked options are concerned, one should be more mindful of his risk than of this reward.

It was also shown previously that a call bull spread does *not* widen out much if the underlying stock makes a quick upward move. The spread won't widen out to its maximum profit potential until expiration draws nigh or the stock is well above the upper strike in the spread. This scenario also does not bode well for the ratio call spread. Suppose that the underlying stock suddenly jumps upward and implied volatility increases at the same time. That combination is seen quite frequently, especially if the stock were previously “dull” or if there is some sort of active corporate (takeover) rumor. The call ratio spread will fare miserably under these conditions, because the increase in stock price certainly harms the naked call position and the bull spread is not widening out much to compensate for it. In addition, the increase in implied volatility is working against both components.

The same sort of thing happens with put ratio spreads. They are really the combination of a put bear spread plus some additional naked put options. If the underlying falls in price, while implied volatility increases—a very common occurrence in all markets—then the put ratio spread will fare poorly. In fact, implied volatility sometimes *explodes* if the underlying falls very rapidly (crashes), so the ratio put spreader should clearly assess his risk in this light.

In summary, a trader utilizing ratio spread strategies should clearly understand and attempt to analyze the risks of an increase in implied volatility. This includes not only assessing the vega risk of the spread, but also using a probability calculator with some inflated volatility estimates to see just what the chances are of the spread getting into real trouble.

BACKSPREADS

A call backspread is merely the opposite of a call ratio spread. Thus, any of the earlier commentary about how an increase in implied volatility is detrimental to a ratio spread can be reversed when discussing the backspread. An increase in implied volatility will be beneficial to a backspread strategy, while a decrease in implied volatility will be slightly harmful to the spread. However, since risk is limited in a backspread, such a decrease in

implied volatility would not have catastrophic consequences unless one had overcommitted his funds to one position.

SUMMARY

In general, one can always determine the exposure of his position to volatility by computing the vega of this position. However, it is also useful for a strategist to have some general feeling for how implied volatility will affect his positions and strategies. Thus, this chapter was designed to point out the most common effects that changes in implied volatility will have on the basic types of option strategies. Once one has a feeling for his exposure to volatility, he can then assess whether an adverse volatility movement is likely. For example, if an increase in implied volatility would be harmful, and the strategist sees that current levels of implied volatility are quite low in comparison to historical norms, then perhaps he should remove or adjust the position.

Volatility and the price of the underlying are the two major components affecting profitability for most option positions. Time decay is only most pertinent as expiration approaches. Yet, many traders concentrate greatly on potential price movements of the underlying, often while ignoring what changes in implied volatility could do. That is a mistake, for the most knowledgeable option traders plan for volatility risk at all times. Understanding and handling that risk can have a positive effect on an option trader's profits.